

✓ Palmer Penguins Statistical Analysis

This notebook contains a comprehensive statistical analysis of the Palmer Penguins dataset using R.

```
# 1. Setup and Package Installation
if(!require(palmerpenguins)) install.packages("palmerpenguins")
if(!require(dplyr)) install.packages("dplyr")
if(!require(ggplot2)) install.packages("ggplot2")
if(!require(e1071)) install.packages("e1071")
if(!require(effsize)) install.packages("effsize")
if(!require(car)) install.packages("car")
```

```
library(palmerpenguins)
library(dplyr)
library(ggplot2)
library(e1071)
library(effsize)
library(car)
```

```
# Load data and omit NAs
data("penguins")
df <- na.omit(penguins)
```

Loading required package: e1071

Warning message in library(package, lib.loc = lib.loc, character.only = TRUE, logical.return = TRUE, :
"there is no package called 'e1071'"
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependency 'proxy'

Loading required package: effsize

Warning message in library(package, lib.loc = lib.loc, character.only = TRUE, logical.return = TRUE, :
"there is no package called 'effsize'"
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Loading required package: car

Warning message in library(package, lib.loc = lib.loc, character.only = TRUE, logical.return = TRUE, :
"there is no package called 'car'"
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

also installing the dependencies 'colorspace', 'fracdiff', 'lmtest', 'timeDate', 'urca', 'zoo', 'RcppArmadillo', 'cowplot'

Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

penguins, penguins_raw

Attaching package: 'e1071'

The following object is masked from 'package:ggplot2':

element

Loading required package: carData

Attaching package: 'car'

The following object is masked from 'package:dplyr':

recode

Task 1: Descriptive Statistical Analysis

```
calc_stats <- function(x) {
  c(
    Mean = mean(x),
    Median = median(x),
    Min = min(x),
    Max = max(x),
    Var = var(x),
    SD = sd(x),
    Q1 = quantile(x, 0.25),
    Q3 = quantile(x, 0.75),
    IQR = IQR(x),
    Skewness = skewness(x),
    Kurtosis = kurtosis(x)
  )
}

# Overall descriptive statistics for body mass
overall_stats <- calc_stats(df$body_mass_g)
print("Overall Statistics for Body Mass:")
print(overall_stats)

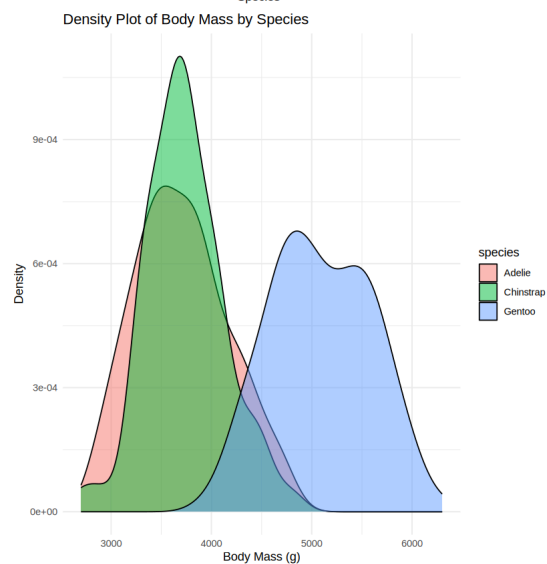
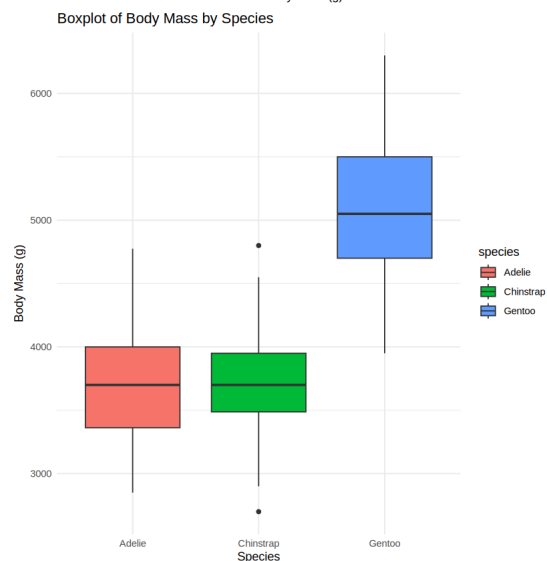
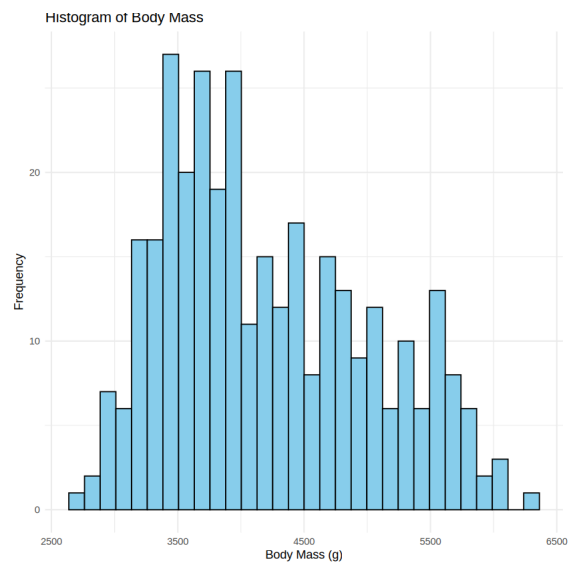
# Species-wise descriptive statistics
species_stats <- df %>%
  group_by(species) %>%
  summarise(
    Mean = mean(body_mass_g),
    Median = median(body_mass_g),
    Min = min(body_mass_g),
    Max = max(body_mass_g),
    Var = var(body_mass_g),
    SD = sd(body_mass_g),
    Q1 = quantile(body_mass_g, 0.25),
    Q3 = quantile(body_mass_g, 0.75),
    IQR = IQR(body_mass_g),
    Skewness = skewness(body_mass_g),
    Kurtosis = kurtosis(body_mass_g)
  )
print("Species-wise Statistics for Body Mass:")
print(species_stats)
```

```
[1] "Overall Statistics for Body Mass:"
      Mean      Median      Min      Max      Var
4.207057e+03 4.050000e+03 2.700000e+03 6.300000e+03 6.483725e+05
      SD      Q1.25%      Q3.75%      IQR      Skewness
8.052158e+02 3.550000e+03 4.775000e+03 1.225000e+03 4.680001e-01
      Kurtosis
-7.540362e-01
[1] "Species-wise Statistics for Body Mass:"
# A tibble: 3 × 12
  species Mean Median Min Max Var SD Q1 Q3 IQR Skewness
<fct>    <dbl> <dbl> <int> <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Adelie 3706. 3700 2850 4775 210332. 459. 3362. 4000 638. 0.274
2 Chinstrap 3733. 3700 2700 4800 147713. 384. 3488. 3950 462. 0.237
3 Gentoo 5092. 5050 3950 6300 251478. 501. 4700 5500 800 0.0444
# 1 more variable: Kurtosis <dbl>
```

```
# Visualizations for Task 1
# Histogram
ggplot(df, aes(x = body_mass_g)) +
  geom_histogram(fill = "skyblue", color = "black", bins = 30) +
  theme_minimal() +
  labs(title = "Histogram of Body Mass", x = "Body Mass (g)", y = "Frequency")

# Boxplot
ggplot(df, aes(x = species, y = body_mass_g, fill = species)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Boxplot of Body Mass by Species", x = "Species", y = "Body Mass (g)")

# Density plot
ggplot(df, aes(x = body_mass_g, fill = species)) +
  geom_density(alpha = 0.5) +
  theme_minimal() +
  labs(title = "Density Plot of Body Mass by Species", x = "Body Mass (g)", y = "Density")
```



Task 2: Hypothesis Testing (Male vs Female)

```
male_mass <- df$body_mass_g[df$sex == "male"]
female_mass <- df$body_mass_g[df$sex == "female"]

# Normality check
shapiro_male <- shapiro.test(male_mass)
shapiro_female <- shapiro.test(female_mass)
print("Shapiro-Wilk test (Male):")
print(shapiro_male)
print("Shapiro-Wilk test (Female):")
print(shapiro_female)
```

```
# QQ Plots
qqnorm(male_mass, main = "QQ Plot - Male Body Mass")
qqline(male_mass, col = "red")

qqnorm(female_mass, main = "QQ Plot - Female Body Mass")
qqline(female_mass, col = "red")

# Independent Two-Sample t-test
ttest_res <- t.test(body_mass_g ~ sex, data = df)
print("T-test Result:")
print(ttest_res)

# Effect size (Cohen's d)
cohen_d <- cohen.d(body_mass_g ~ sex, data = df)
print("Cohen's d:")
print(cohen_d)
```

```
[1] "Shapiro-Wilk test (Male):"
```

Shapiro-Wilk normality test

data: male_mass

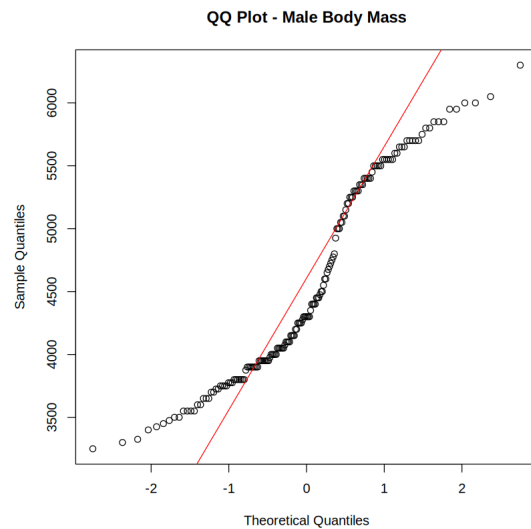
W = 0.92504, p-value = 1.227e-07

```
[1] "Shapiro-Wilk test (Female):"
```

Shapiro-Wilk normality test

data: female_mass

W = 0.91931, p-value = 6.155e-08



```
[1] "T-test Result:"
```

Welch Two Sample t-test

data: body_mass_g by sex

t = -8.5545, df = 323.9, p-value = 4.794e-16

alternative hypothesis: true difference in means between group female and group male is not equal to 0

95 percent confidence interval:

-840.5783 -526.2453

sample estimates:

mean in group female	mean in group male
3862.273	4545.685

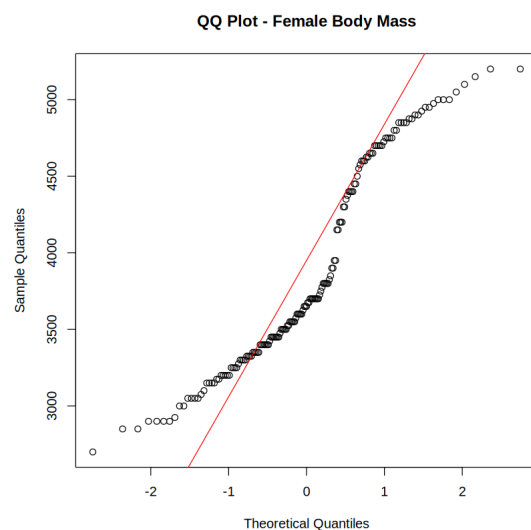
```
[1] "Cohen's d:"
```

Cohen's d

d estimate: -0.9362048 (large)

95 percent confidence interval:

lower	upper
-1.1633158	-0.7090938



Task 3: One-Way ANOVA

```
# Check homogeneity of variance
levene_res <- leveneTest(body_mass_g ~ species, data = df)
print("Levene's Test:")
print(levene_res)

# One-way ANOVA
anova_res <- aov(body_mass_g ~ species, data = df)
print("One-Way ANOVA Summary:")
summary(anova_res)

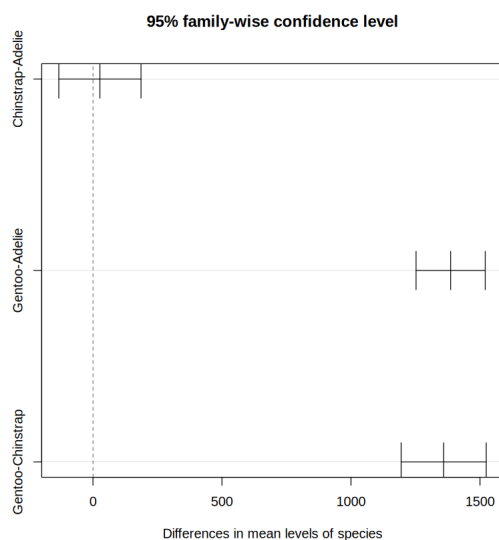
# Tukey HSD Post-hoc
tukey_res <- TukeyHSD(anova_res)
print("Tukey HSD Post-hoc Test:")
print(tukey_res)

# Group Comparison Plot
plot(tukey_res)
```

```
[1] "Levene's Test:"
Levene's Test for Homogeneity of Variance (center = median)
      Df F value    Pr(>F)
group  2  5.1349 0.006367 **
330
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[1] "One-Way ANOVA Summary:"
      Df    Sum Sq Mean Sq F value Pr(>F)
species  2 145190219 72595110   341.9 <2e-16 ***
Residuals 330  70069447  212332
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
[1] "Tukey HSD Post-hoc Test:"
      Tukey multiple comparisons of means
      95% family-wise confidence level

Fit: aov(formula = body_mass_g ~ species, data = df)

$species
      diff      lwr      upr    p adj
Chinstrap-Adelie  26.92385 -132.3528  186.2005 0.916431
Gentoo-Adelie    1386.27259 1252.2897 1520.2554 0.000000
Gentoo-Chinstrap 1359.34874 1194.4304 1524.2671 0.000000
```



Task 4: Non-Parametric Analysis

```
kruskal_res <- kruskal.test(body_mass_g ~ species, data = df)
print("Kruskal-Wallis Test:")
print(kruskal_res)
```

```
[1] "Kruskal-Wallis Test:"

      Kruskal-Wallis rank sum test

data:  body_mass_g by species
```

Kruskal-Wallis chi-squared = 212.09, df = 2, p-value < 2.2e-16

Task 5: Two-Way ANOVA

```

twoway_anova <- aov(body_mass_g ~ species * sex, data = df)
print("Two-Way ANOVA Summary:")
summary(twoway_anova)

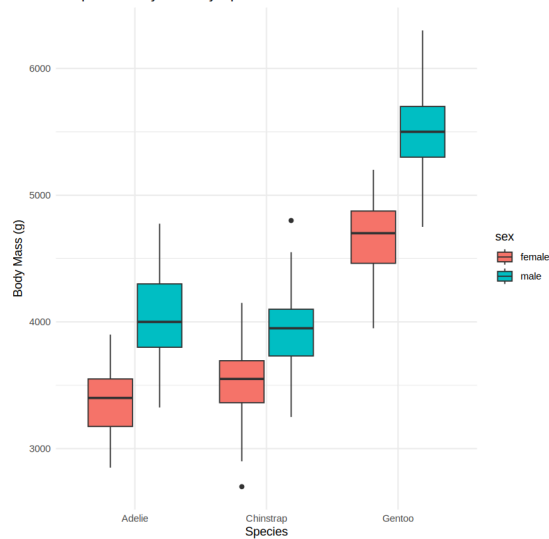
# Sex-wise boxplot
ggplot(df, aes(x = species, y = body_mass_g, fill = sex)) +
  geom_boxplot() +
  theme_minimal() +
  labs(title = "Boxplot of Body Mass by Species and Sex", x = "Species", y = "Body Mass (g)")

```

```

[1] "Two-Way ANOVA Summary:"
      Df Sum Sq Mean Sq F value    Pr(>F)
species  2 145190219 72595110  758.358 < 2e-16 ***
sex       1  37090262  37090262  387.460 < 2e-16 ***
species:sex  2  1676557   838278   8.757 0.000197 ***
Residuals 327  31302628   95727
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Boxplot of Body Mass by Species and Sex

```



Task 6: Additional Analysis (Flipper Length)

```

anova_flipper <- aov(flipper_length_mm ~ species, data = df)
print("One-Way ANOVA (Flipper Length):")
summary(anova_flipper)

```